Wendy Q. Sun

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EDUCATION

Massachusetts Institute of Technology — Cambridge, MA
Double major: B.S. in Physics — 2022 – 2026
B.S. in Artificial Intelligence & Decision Making
Minor in Mathematics
GPA : 4.9/5.0

HONORS

Phi Beta Kappa Honor Society (top 10% of the graduating class) 2026
Sigma Pi Sigma Physics and Astronomy Honor Society 2026
Tau Beta Pi Engineering Honor Society 2026

PUBLICATIONS

3 primary author papers, ADS library. 16 total papers, ADS library.
† marks equal contribution.

Primary Author Papers

3. **Wendy Q. Sun**, Rohan P. Naidu, Hanpu Liu, Anna de Graaff, Jenny E. Greene, Jorryt Matthee, et al., *Overmassive No More: The Case for Little Red Dots Hosting Black Hole Seeds as Massive as Single Supermassive Stars*, submitted to OJAp [8 citations].
2. **Wendy Q. Sun**, Rohan P. Naidu, Jorryt Matthee, Anna de Graaff, et al., *Little Red Dot − Host Galaxy = Black Hole Star: A Gas-Enshrouded Heart at the Center of Every Little Red Dot*, OJAp, 9, 62505, 2026 [87 citations].
1. C. Loughridge†, **Q. Sun**†, et al., *DafnyBench: A Benchmark for Formal Software Verification*, The 4th Workshop on Mathematical Reasoning and AI at NeurIPS, 2024 [113 citations].

Contributing Author Papers

13. G. Limberg et al., *Chemodynamical Analysis of a CNO-Enhanced Ultra Metal-poor Star ([Fe/H] < −4): Insights into Early Enrichment by Faint Population III Supernovae*, submitted to AAS journals.
12. C. Ashall et al., *Can Little Red Dots Contribute To The Early Universe Cosmic Dust Budget?*, submitted to ApJ.
11. R. P. Naidu et al., *A gas-enshrouded and gas-reddened black hole at cosmic dawn*, Nature, 656, 329, 2026.
10. Y. Ishikawa et al., *Spatial decomposition of Little Red Dots with JWST/NIRSpec IFU into broad-line red cores and narrow-line blue host galaxies*, arXiv.
9. R. P. Naidu et al., *Little Red Dots as Intermediate Mass, Super-Eddington Engines: Insights from Type IIn Supernovae and The 1837–1856 Great Eruption of η Carinae*, submitted to OJAp.
8. A. Weibel, R. P. Naidu, et al., *Black Hole Stars Across the Universe: Identifying Central Engine Dominated Little Red Dots at $z \sim 1.5 - 9.5$*, submitted to OJAp.
7. B. Wang et al., *Water absorption confirms cool atmospheres in two little red dots*, arXiv.
6. A. Torralba et al., *A Black Hole Star at Cosmic Noon: Extreme Balmer break, photospheric continuum, and broad absorption by thick winds in a Little Red Dot at $z = 1.7$*, ApJL, 1005, 37, 2026.
5. Z. Liu, R. P. Naidu, A. Secunda, et al., *How I Wonder What You Are – JWST's Little Red Dots do not TWINKLE*, submitted to ApJ.

4. T. Li, **Q. Sun**, et al., *Fractal Generative Models*, Transactions of Machine Learning Research (TMLR), 2025.
3. J. Colaço Carr, **Q. Sun**, et al., *Focused Skill Discovery: Learning to Control Specific State Variables while Minimizing Side Effects*, The 2nd Reinforcement Learning Conference (RLC), 2025.
2. S. Casper, C. Ezell, et al., *Black-Box Access is Insufficient for Rigorous AI Audits*, The ACM Conference on Fairness, Accountability, and Transparency (FAccT), 2024.
1. W. Gurnee et al., *Universal Neurons in GPT2 Language Models*, Transactions of Machine Learning Research (TMLR), 2024.

PRESENTATIONS

ISSI Bern, *LRD − Host Galaxy = BH*: A Gas-Enshrouded Heart at the Center of Every LRD*
Meeting, Feb. 2026

RESEARCH EXPERIENCE

Weighing Central Engines of Little Red Dots — Honolulu, HI
Institute for Astronomy, University of Hawai'i at Mānoa • *Advisor: Rohan Naidu* — July 2026 – present
· Estimated the masses of central engines in little red dots by fitting stellar atmosphere models, finding masses consistent with the engines arising from supermassive stars.

Host Galaxy Separation for Little Red Dots — Cambridge, MA
MIT Kavli Institute • *Advisor: Rohan Naidu* — July 2025 – May 2026
· Separated the spectra of 90+ little red dots into central engine and host galaxy components, finding population-level evidence that their central engines are black hole stars.

Fractal Generative Models — Cambridge, MA
MIT CSAIL, He Vision Group • *Advisors: Tianhong Li, Kaiming He* — Nov. 2024 – Feb. 2025
· Contributed to developing and testing a novel AI generative model with fractal-like structure that nests smaller generative models within larger ones.

DafnyBench: A Benchmark for Formal Software Verification — Cambridge, MA
NSF IAIFI, Tegmark AI Safety Group • *Advisor: Max Tegmark* — Feb. – Nov. 2024
· Built the then-largest software verification benchmark (DafnyBench, 750+ programs) for evaluating how well AI models can verify code beyond just generating it.
· Built a VS Code extension that uses AI models to verify code (Dafny Autopilot, 400+ installs).

Focused Skill Discovery for Reinforcement Learning — Berkeley, CA
Center for Human-Compatible AI, UC Berkeley • *Advisor: Cam Allen* — June – Aug. 2024
· Implemented and evaluated algorithms that keep AI agents from causing harmful side effects.

SKILLS

Astronomy software Cloudy, Prospector, dynesty

RELEVANT COURSEWORK

Physics Astrophysics I (grad), General Relativity (grad), Statistical Mechanics I (grad), Relativistic Quantum Field Theory I (grad), Electromagnetic Theory (grad)
Mathematics Real Analysis, Numerical Analysis, Theory of Computation, Combinatorial Optimization